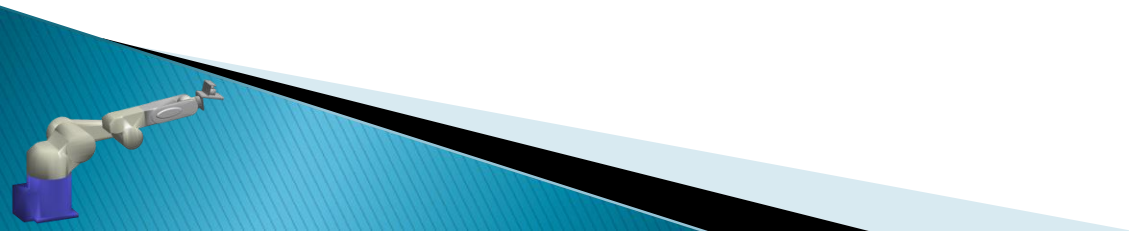


Week 4

Schedule

- ▶ Collaborative exercise
- ▶ Safety documentation
- ▶ Review & discussion of Lab 3 (5 mins)
- ▶ Introduction to Week 4 and Lab 4 (15 mins)
- ▶ Reminder about Quiz 2 (5 mins)
- ▶ Collaboratively work on Lab 4 (≈ 1.5 hours)



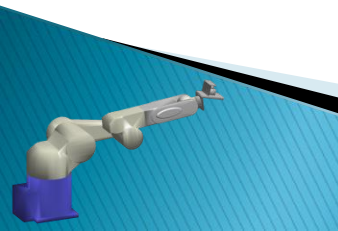
Activity Introduction

We are conducting a research study to define and differentiate collaborative robots (cobots) using Group Concept Mapping. Your budding expertise in robotics would be invaluable. We aim to gather diverse perspectives to shape future cobot frameworks.

- ▶ **Human-Robot Collaboration Discussion**
- ▶ Each table (~8 people) will receive **one question**.
- ▶ Discuss together and capture your ideas on sticky notes.
- ▶ Paste your notes onto the sheet provided.
- ▶ The goal is to **collect ideas**, not find a single correct answer.
- ▶ 💡 Think beyond robot arms: E.g., Mobile robots, exoskeletons, drones, social robots, cobots in logistics, healthcare, mining, agriculture, and more.
- ▶ Each table will work on one of these:
 - What does it mean for a human to collaborate with a robot or vice versa?
 - In terms of human-robot collaboration, what makes a robot collaborative for you?
 - Can you describe a few key applications that demonstrate different collaborative features or qualities of robots?
- ▶ Please write neatly, and have a tutor mark your participation and photograph your table's work

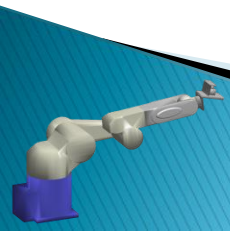
Sector \ Categories	Manufacturing	Logistics	Service + Hospitality	Healthcare + Elderly care	Defence	Agriculture
Academia						
Companies						
Associations						
Regulators (e.g. WHS)						
Policy makers						

Worth considering



Census date 28/8

- **Financial Liability:** You become liable for enrolled subjects.
- **Withdrawal Deadline:** Last day to withdraw without financial or academic penalty.
- **HELP Debt:** Incur a HELP debt if eligible and no upfront payment is made.
- **Commonwealth Support:** Last day to submit forms, provide TFN, and make upfront payments for Commonwealth supported places.
- **Advice:** If you're feeling unsure of a subject choice, or your work, life or family commitments are getting on top of you and you don't feel like you can continue with your current number of subjects, now is the best time to withdraw
- <https://www.uts.edu.au/current-students/managing-your-course/important-dates/census-date>

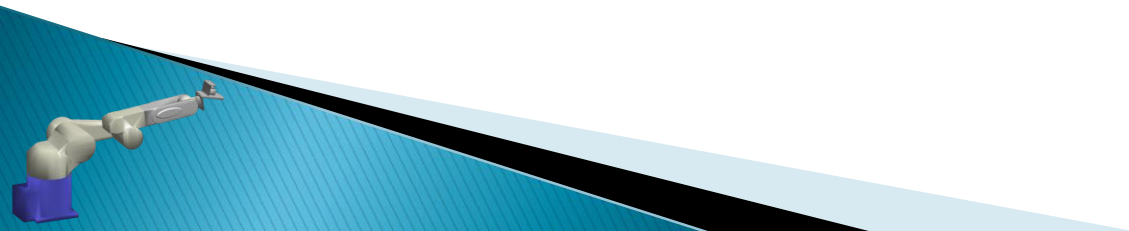


Lab Drop-In Sessions (with a tutor)

- ▶ CB11.10.403 in W4,5,6,10,11,12

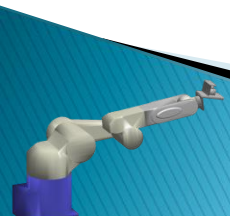
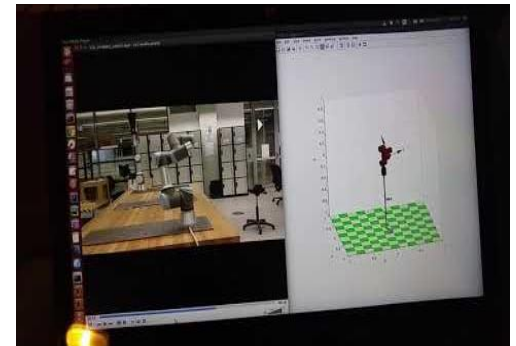
Timeslots and dates are:

- 9–5pm on ~~18/08~~, 25/08, 01/09
- 9–3pm on 9/10, 16/10, 23/10



Lab Assignment 1

- ▶ Ensure you have done your Mechatronics Lab induction
 - "FEIT CB11 Mechatronics Lab Induction" in RapidGlobal: <https://inductions.uts.edu.au/>
- ▶ Submit to Canvas get it checked off
 - Completed and signed Risk Assessment
 - Completed and signed Safe Work Method Statement (SWMS)
 - video evidence of a correct robot model





<https://www.youtube.com/watch?v=4agh-pml8Go>



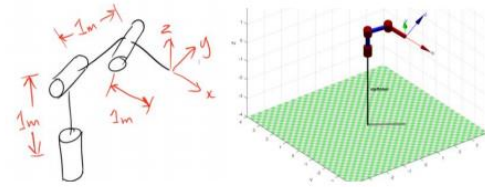
<https://www.youtube.com/watch?v=9yElmIBrZAQ>

Lab 3 Review – Q1

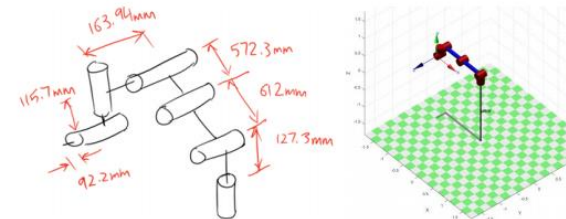
1. Who watched the solution video?
2. Derive DH parameters for each manipulator
 - i. Be familiar with the variables: θ , d , a , α
 - ii. Use the teach function, is it what you expect?
 - iii. Obtain the current joint angles
 - iv. Determine the forward kinematics transformation matrix
 - v. Given some transformation T , determine the joint angles required to satisfy T
 - vi. What does the Jacobian tell you about the end-effector velocity given some joint configuration?

ROBOT 1: 3-Link 3D Robot

Note the orientation of the end-effector frame.

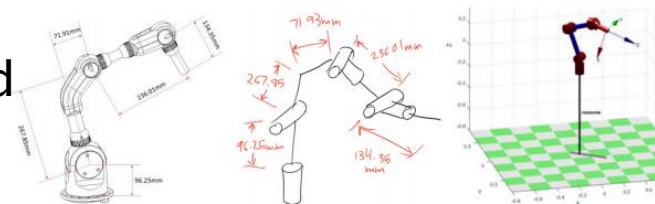


ROBOT 2: AAN-BOT (Assistance As Needed BOT) (6DOF)^{[1][2]}



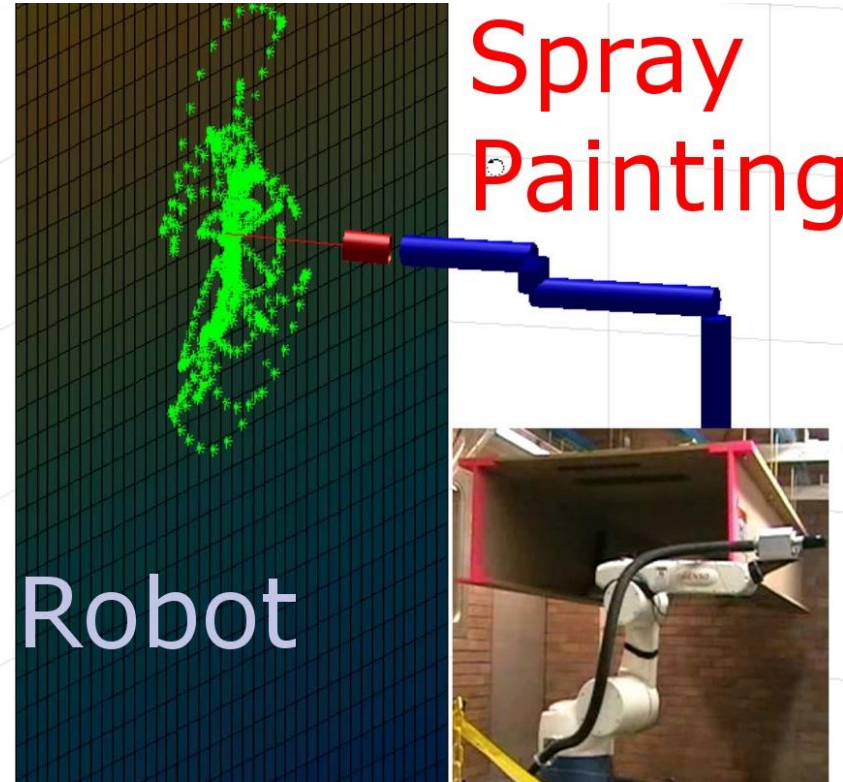
ROBOT 3: Submerged Pile Inspection Robot (SPIR)^[3]

Note: this arm has 2DOF joints.



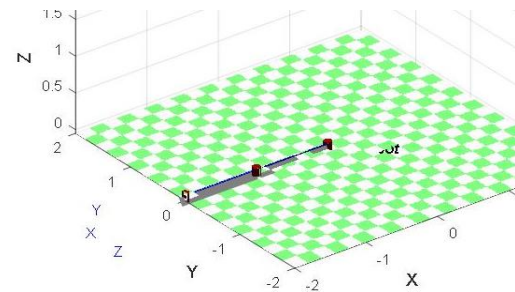
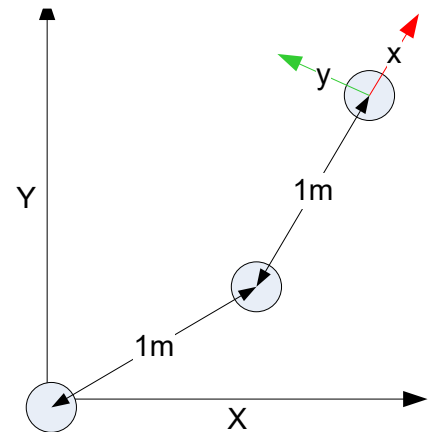
Lab 3 Review – Q2–Q3

1. Derive DH parameters for the Denso VM-6083D-W
2. Sample the joint angles of the manipulator then determine the forward transformation matrix at each joint configuration
3. Using the list of XYZ locations obtained from the forward transformation matrices plot the workspace of the manipulator
4. Line Plane Intersection



Introduction to Lab 4 – Q1

- ▶ Look at Lab 4 – Trajectory Interpolation Reference Guide.pdf
- ▶ Draw with 3DOF Planar Arm so the Z axis faces down ways
- ▶ Move robot with “teach”. Observe how there’s no way to affect the Z, roll or pitch values, no matter what joint value you choose.
- ▶ Use ikine and mask out the impossible-to-alter values
- ▶ Go through a loop to draw a “straight” line and observe its straightness
- ▶ Trace out a circle using ikine

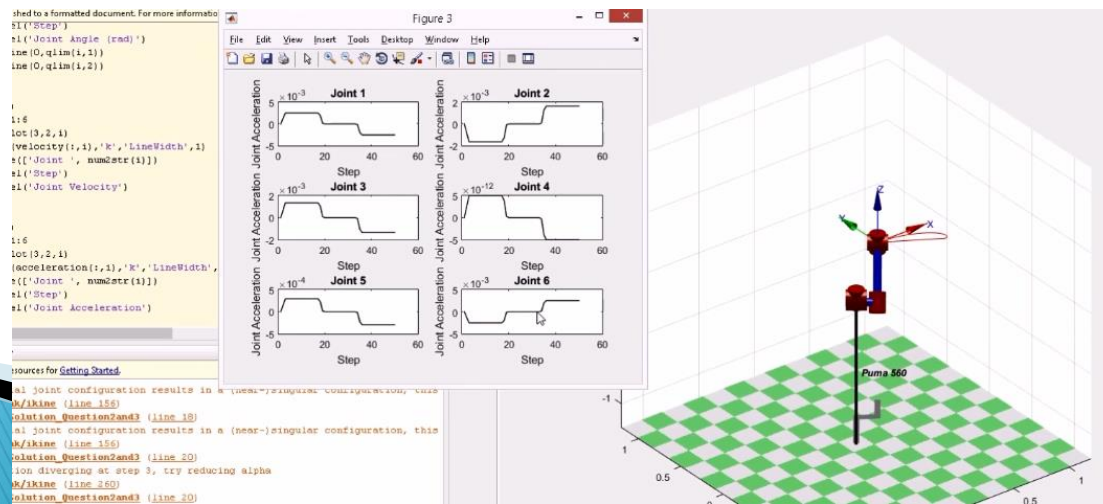


Introduction to Lab 4 – Q2

- ▶ Inverse Kinematics to Determine Joint Angles of Puma 560
- ▶ Solve inverse kinematics to get the required joint angles
- ▶ This is similar to the assignment question

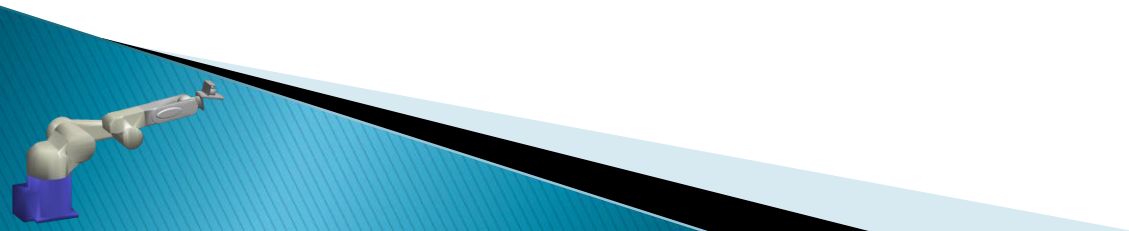
Introduction to Lab 4 – Q3

- ▶ Interpolation: Get from A to B using puma560
- ▶ Download Trajectory Interpolation Reference Guide
- ▶ Generate a matrix of interpolated joint angles with 50 steps between $q1$ and $q2$. Do steps 3.4 to 3.10 for each of the two methods:
 - Quintic Polynomial Profile
 - Trapezoidal Velocity Profile



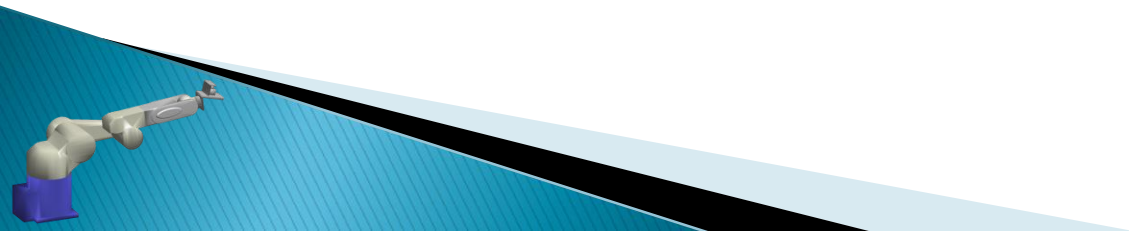
Overview of Week 5: Quiz 2

- ▶ Two restricted open book quizzes worth 5% in total
- ▶ Individual quiz
 - worth 5%
 - will start at the same time as the lab
 - under 10 questions:
 - a few multiple choice about theory (focus on Modules 3 & 4),
 - a couple from Quiz 1 (that people did poorly in Week 3)
 - a couple from Lab3/4 Starters/Solutions
 - ~35 minutes
 - No talking or communicating



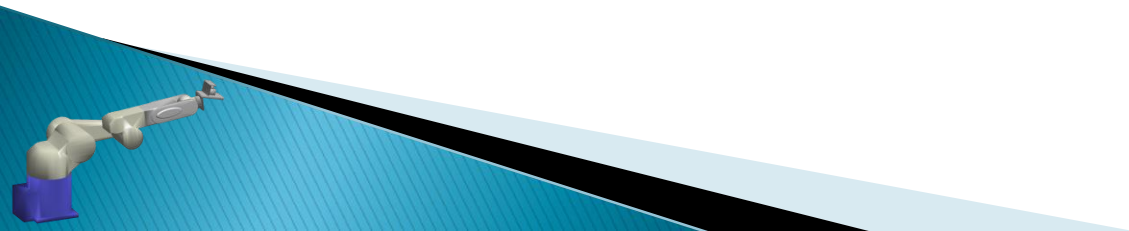
Pre-work before Week 5 lab

- ▶ Completed week 4 lab exercises
- ▶ Review “Week 4 – Trajectory Interpolation Reference Guide.pdf”
- ▶ Ensure you have achieved 80% in Quiz 1
- ▶ Work on Lab Assignment #1 (due in week 6 lab)



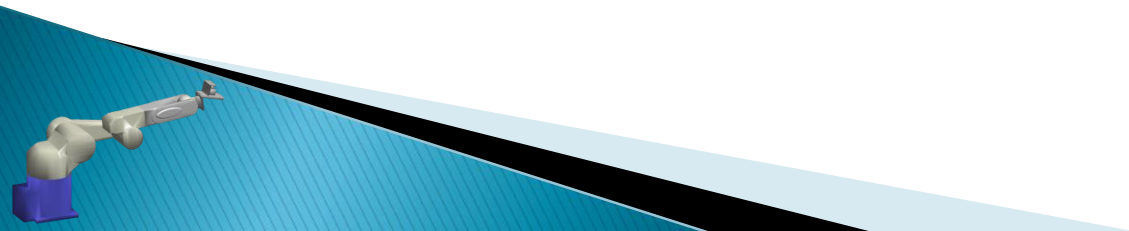
Note/slides from the textbook

- ▶ Textbook readings for week 4:
 - Section 7.4 (pages 152–158): “Trajectories
 - Section 7.3 (pages 146–152): “Inverse Kinematics”
- ▶ Although it is better to read the textbook, some notes (in slide format) have been summarised below



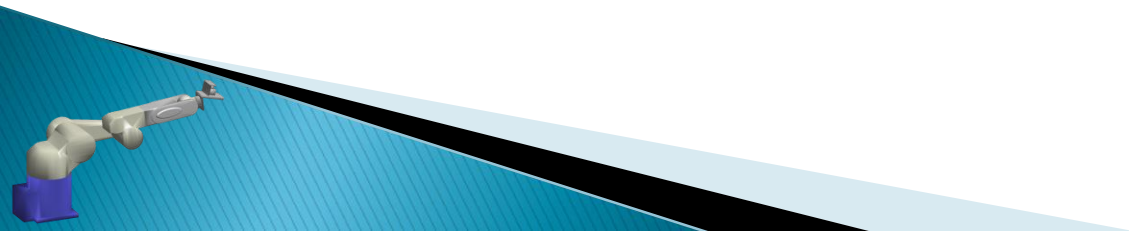
Inverse Kinematics

- ▶ This is the inverse kinematics problem which is written in functional form as:
$$q = K^{-1}(\xi)$$
- ▶ In general this function is not unique and for some classes of manipulator no closed form solution exists, necessitating a numerical solution.



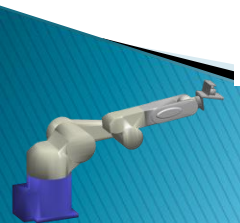
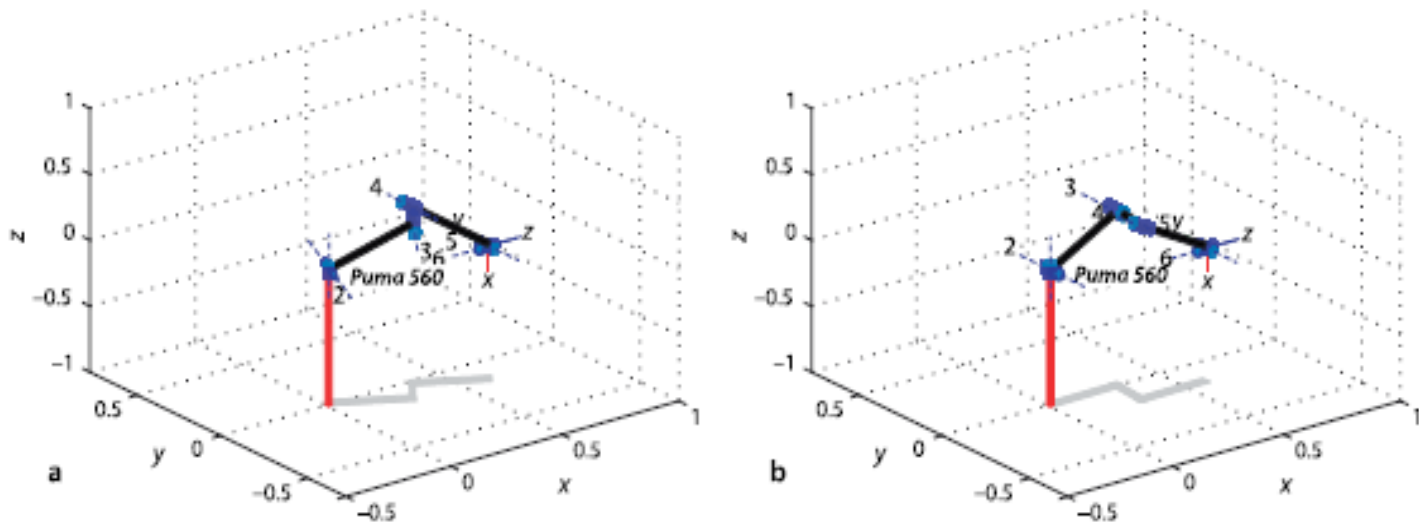
Closed-Form Solution

- ▶ A closed-form solution of a 6-axis robot is that the three wrist axes intersect at a single point.
- ▶ This means that motion of the wrist joints only changes the end-effector orientation, not its translation.



Closed-Form Solution (continued...)

- ▶ **Fig. 7.6.** Two solutions to the inverse kinematic problem, left-handed and right-handed solutions. The shadow shows clearly the two different configurations



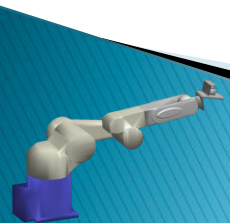
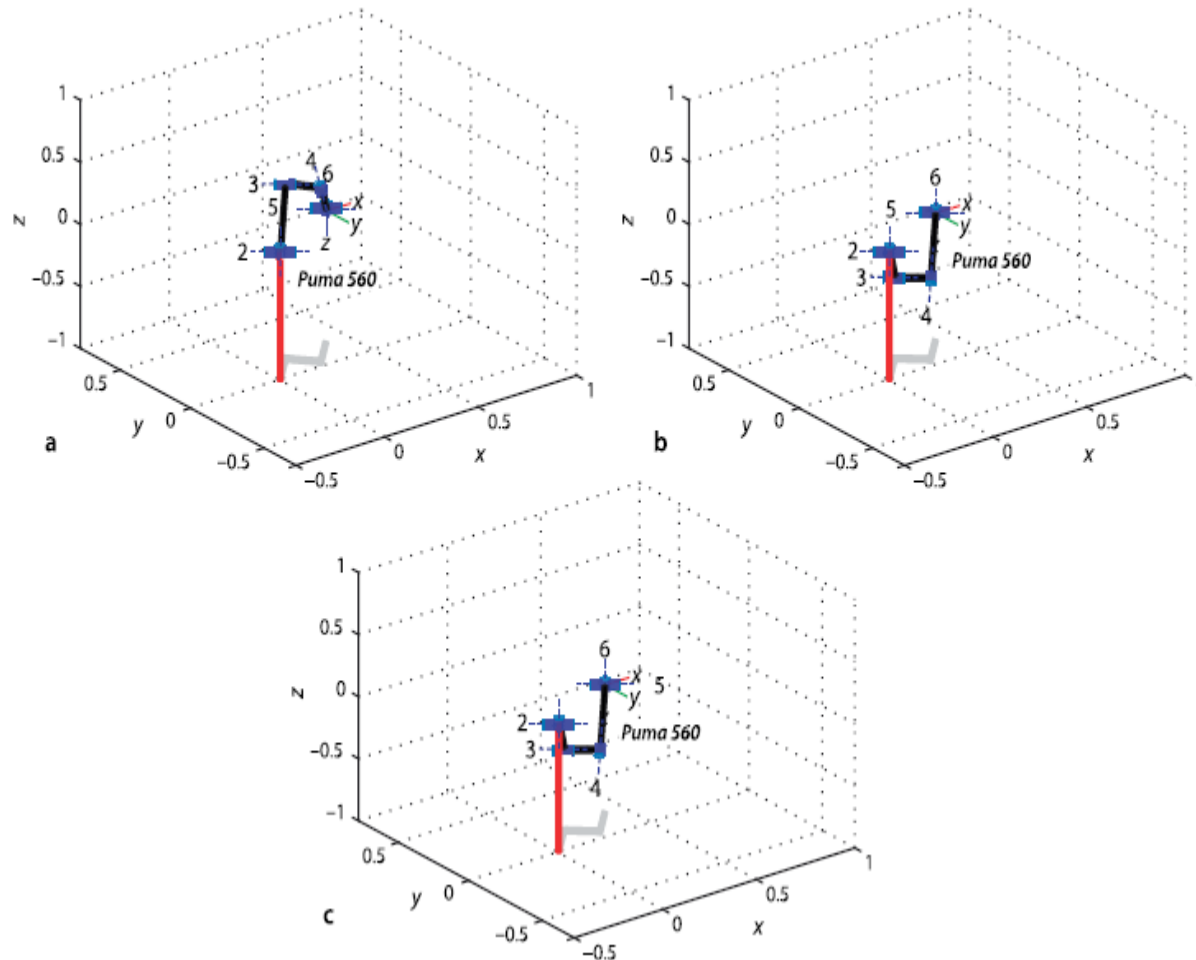
Closed-Form Solution (continued...)

Fig. 7.7. Different configurations of the Puma 560 robot

a) Right-up-noflip;

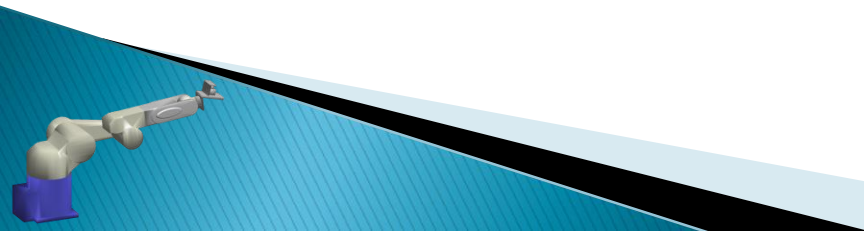
b) right-down-noflip;

c) right-down-flip



Under-Actuated Manipulator

- ▶ An under-actuated manipulator is one that has fewer than six joints, and is therefore limited in the end-effector poses that it can attain.
- ▶ SCARA robots are a common example.
- ▶ They typically have an $x-y-z-\theta$ **task space**, $\mathcal{T} \subset \mathbb{R}^3 \times \mathbb{S}$ and a **configuration space** $\mathcal{C} \subset \mathbb{S}^3 \times \mathbb{R}$.

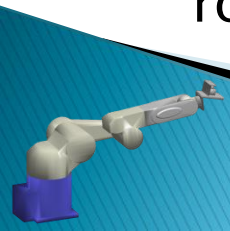


Redundant Manipulator

- ▶ A redundant manipulator is a robot with more than six joints.
- ▶ Six joints is theoretically sufficient to achieve any desired pose in a Cartesian taskspace $\mathcal{T} \subset SE(3)$.
- ▶ The Denavit–Hartenberg (DH) parameters for the base are

Link	θ_i	d_i	a_i	α_1	σ_i
1	0	q_1	0	$-\frac{\pi}{2}$	1
2	$-\frac{\pi}{2}$	q_2	0	$\frac{\pi}{2}$	1

- ▶ The DH notation requires that prismatic joints cause translation in the local z -direction and the base-transform rotates that z -axis into the world x -axis direction.



Redundant Manipulator

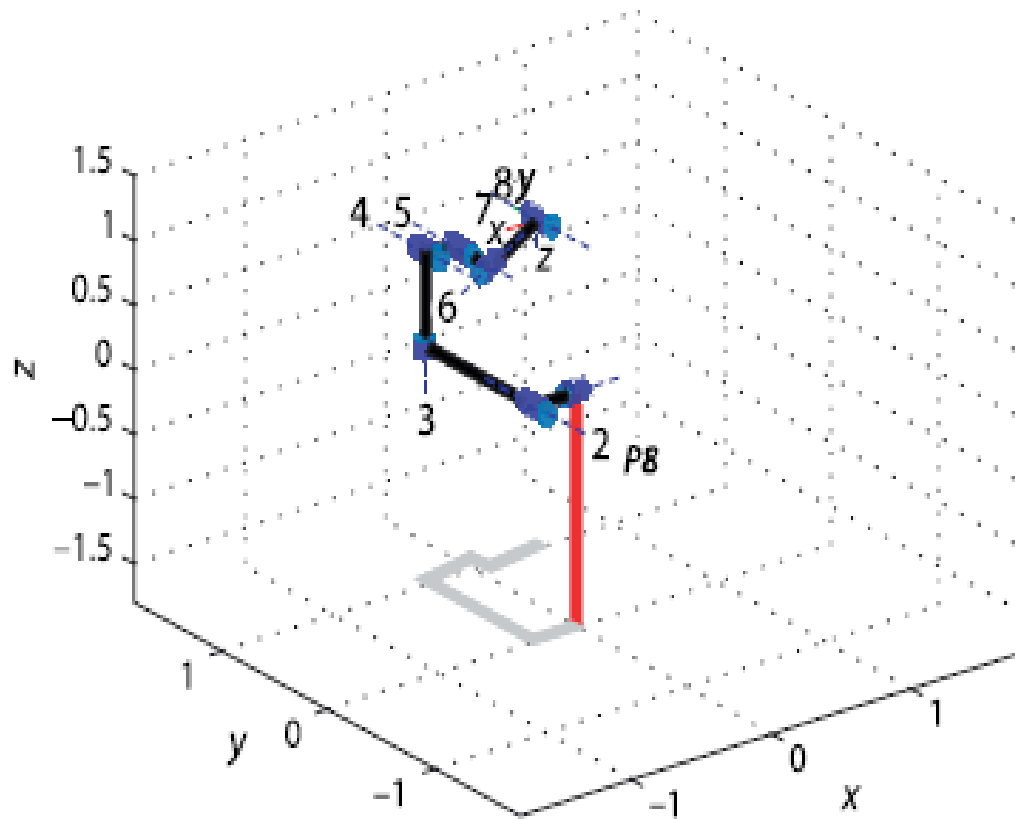
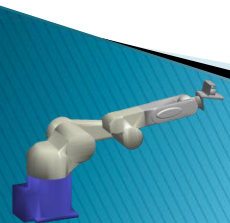
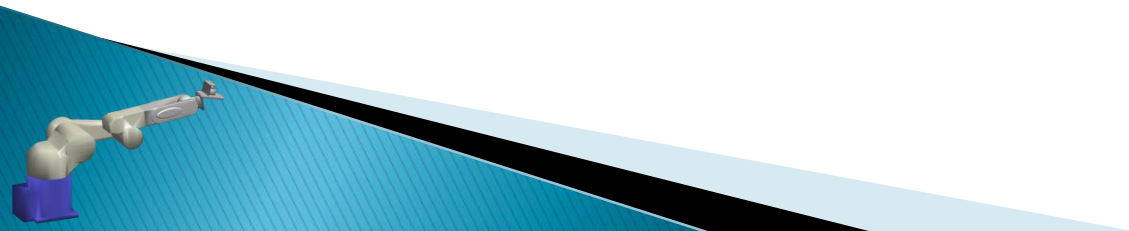


Fig. 7.8. Redundant manipulator



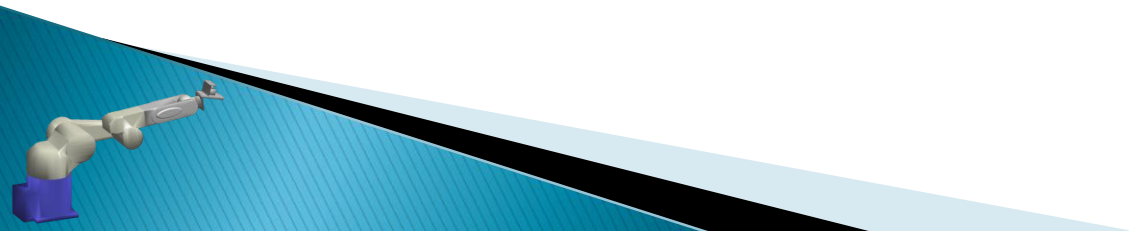
Trajectories

- ▶ One of the most common requirements in robotics is to move the end-effector smoothly from pose A to pose B.
- ▶ Trajectories has two approach:
 - straight lines in joint space
 - straight lines in Cartesian space.



Joint-Space Motion

- ▶ A joint-space trajectory is formed by smoothly interpolating between two configurations q_1 and q_2 .
- ▶ The joint coordinate trajectory is smooth but we do not know how the robot's end-effector will move in Cartesian space.



Joint-Space Motion (continued...)

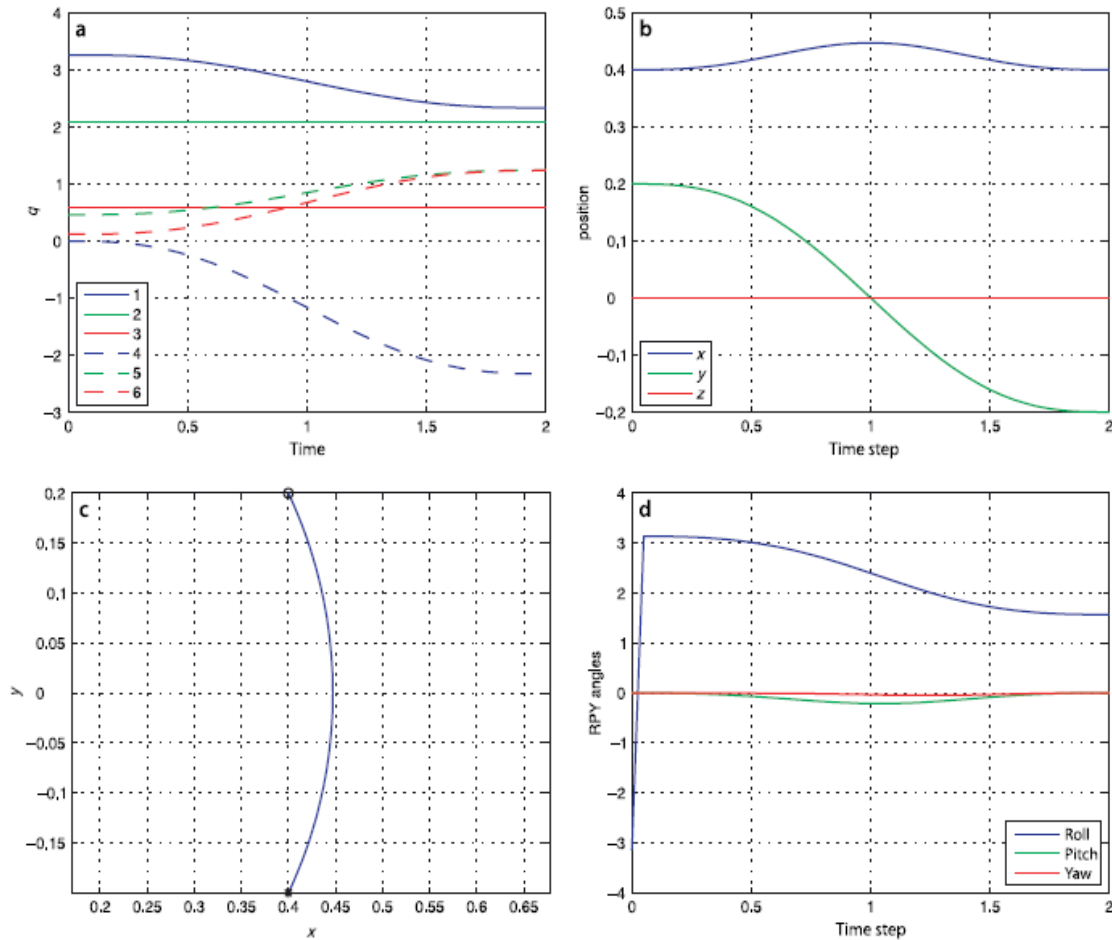
Fig. 7.9. Joint space motion.

a Joint coordinates vs time;

b Cartesian position vs time;

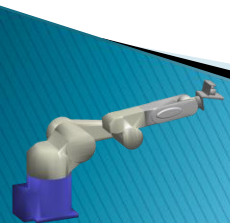
c Cartesian position locus in the xy -plane

d roll-pitch-yaw angles vs time



Cartesian Motion

- ▶ The joint alignment means that the robot has lost one degree of freedom and is now effectively a 5-axis robot.
- ▶ Kinematically we can only solve for the sum $q_4 + q_6$ and there are an infinite number of solutions for q_4 and q_6 that would have the same sum.



Cartesian Motion (continued...)

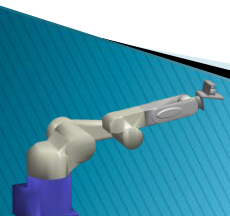
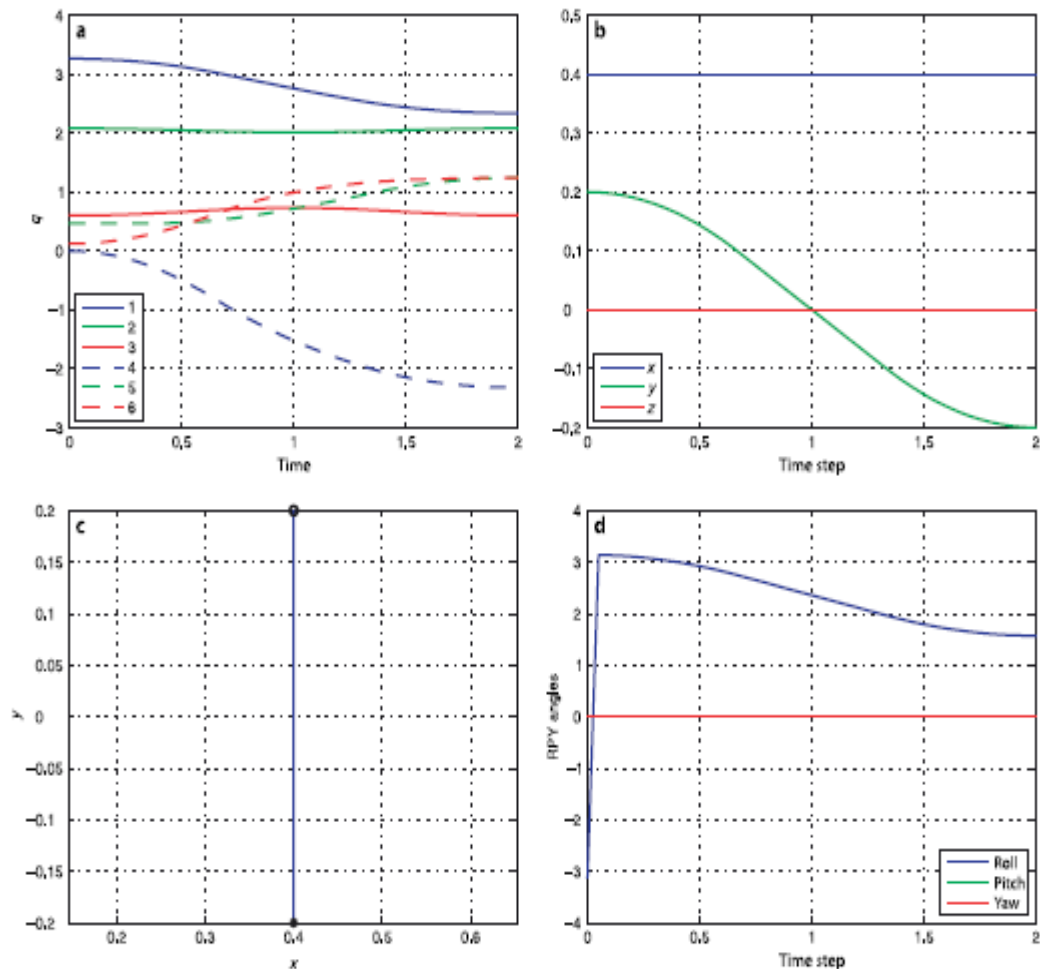
Fig. 7.11. Cartesian motion.

a) Joint coordinates vs time;

b) Cartesian position vs time;

c) Cartesian position locus in the xy -plane;

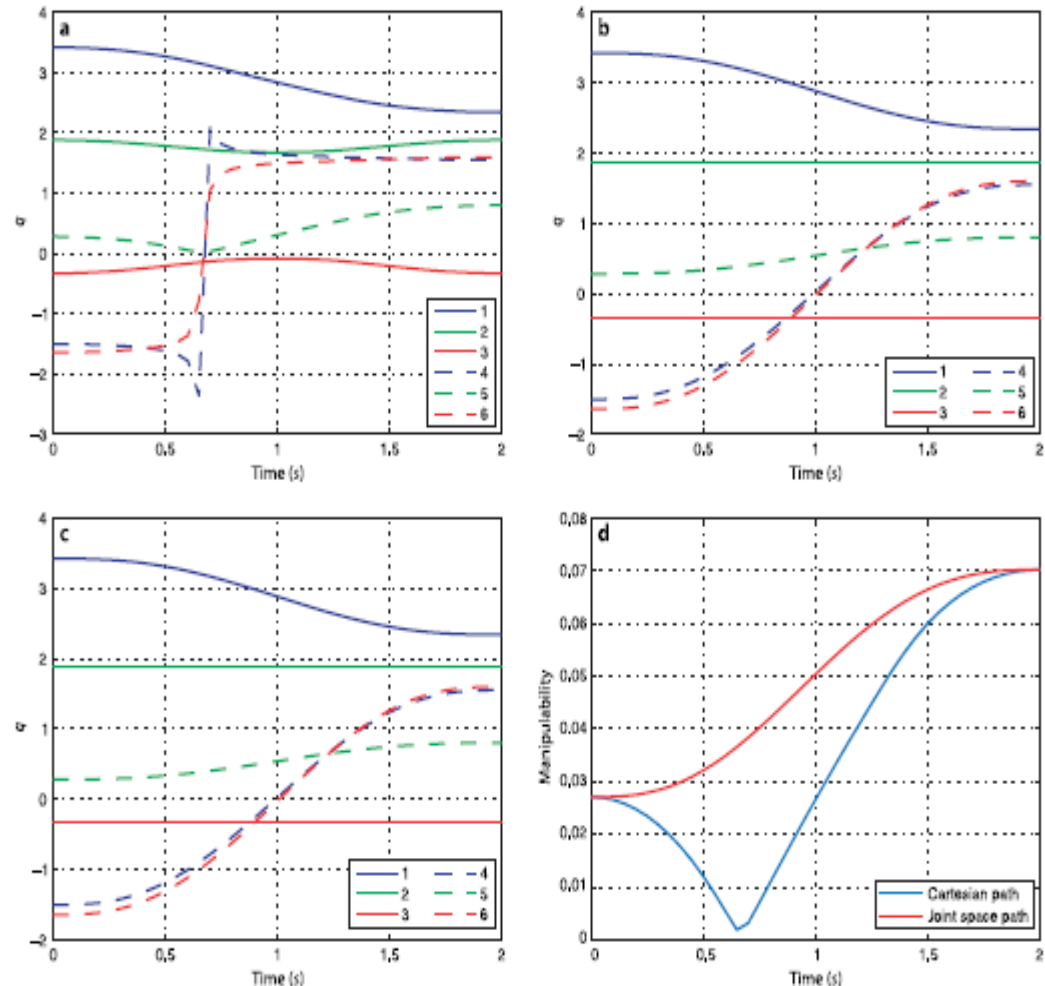
d) roll-pitch-yaw angles vs time



Cartesian Motion (continued...)

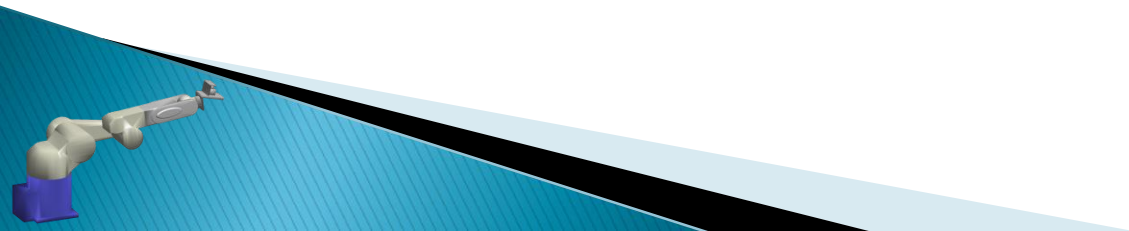
- **Fig. 7.12.** Cartesian motion through a wrist singularity.

- a) Joint coordinates computed by inverse kinematics (ikine6s);
- b) joint coordinates computed by generalized inverse kinematics (ikine);
- c) joint coordinates for joint-space motion;
- d) manipulability



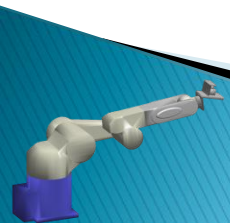
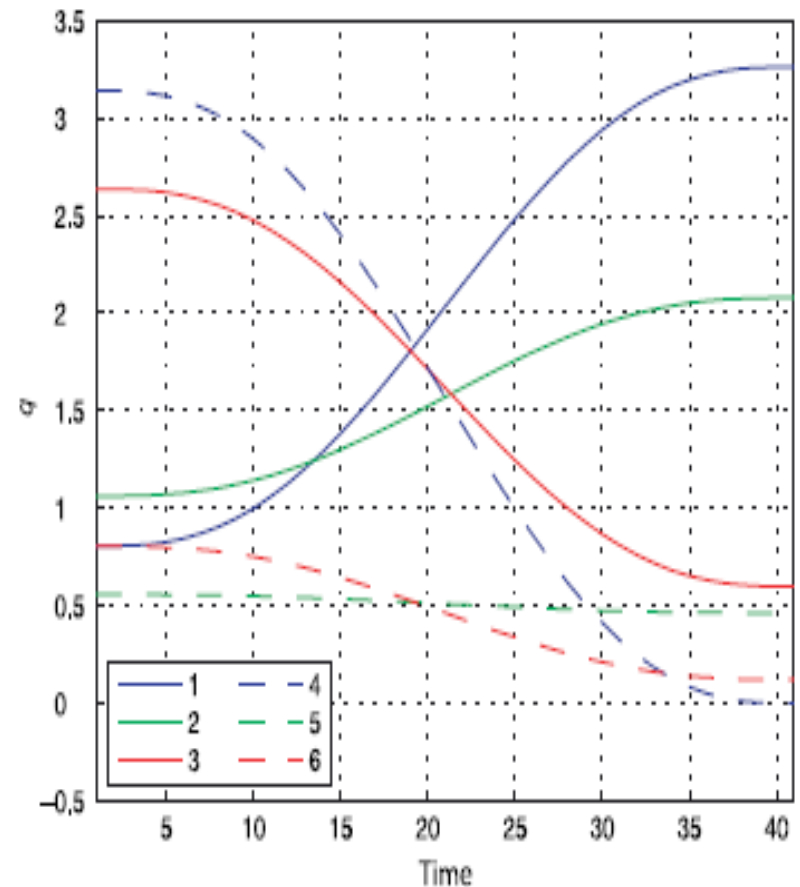
Configuration Change

- ▶ Movement from one configuration to another ultimately results in no change in the end-effector pose since both configuration have the same kinematic solution
- ▶ Therefore we *cannot* create a trajectory in Cartesian space. Instead we must use joint-space motion.



Configuration Change (continued...)

- ▶ **Fig. 7.13.** Joint space motions for configuration change from right-handed to left-handed



References

- ▶ [1] Robotics, Vision and Control. Peter Corke